

Lex can be used to develop your own chatbots. There is also an open source language model called GPT-J, which is similar to OpenAI's GPT-3. Notice that ChatGPT uses GPT-3.5 (used by the free product) and GPT-4 (used by the premium product). Microsoft's search engine called Bing also has features based on GPT-4. Other chatbots available online include Perplexity, Jasper, Chatsonic (Writesonic), Replika, Tome, etc. However, many of these services are not free like ChatGPT. Interestingly, some of these tools even provide the option to rewrite text to bypass AI content detection.

A few programming languages for AI

Now, let us talk about some programming languages that are useful for developing AI and machine learning applications. Before we dive in, we need to address two important questions. First, why are we discussing other programming languages towards the end of this series when we have spent a lot of time learning Python-based techniques? Well, we are doing so because these languages provide benefits such as faster processing, better portability, greater low-level control, and improved scalability for developing AI and machine learning applications. Additionally, we will only focus on programming languages that have characteristics distinct from Python to make this discussion worthwhile.

The second question is: What should you do when you have a strong affinity for a programming language that is not listed among the popular options for developing AI and machine learning applications? Fortunately, times have changed, and no programming language or development community can ignore the growing importance of AI. Almost every programming language has some features to support AI-based

```
# sbcl
This is SBCL 2.0.1.debian, an implementation of ANSI Common Lisp.
More information about SBCL is available at <http://www.sbcl.org/>.

SBCL is free software, provided as is, with absolutely no warranty.
It is mostly in the public domain; some portions are provided under
BSD-style licenses. See the CREDITS and COPYING files in the
distribution for more information.
* (load "add.lisp")
Program Loaded
T
* (add 111 222)
333
```

Figure 1: Execution of the Lisp program

development. For instance, if you prefer JavaScript, a popular language for web development, there are now certain libraries and frameworks available to support AI and machine learning, such as TensorFlow.js, Brain.js, and ConvNetJS. These libraries enable developers to perform machine learning tasks directly in the browser, which is useful for creating interactive applications that rely on real-time predictions.

Now, let us explore some of the key programming languages that are used for developing AI.

Prolog is a logic programming language that is well-suited for certain types of AI applications, such as expert systems and natural language processing. Released in 1971, Prolog is actively used today, though not at the same scale as many other programming languages discussed in this series. For example, one of the languages used to develop IBM Watson is Prolog. You can execute Prolog programs in your system by using a compiler called GNU Prolog. It can be installed in Ubuntu by using the command, '*sudo apt install gprolog*'. Prolog is worth considering as a potential candidate for AI development due to its unique programming paradigm. However, keep in mind that this language is used sporadically, and the chances of coming into contact with a professional Prolog programmer are slim.

Lisp is another historically (being the second oldest surviving high-level programming language) and practically relevant programming language for AI development. Although Lisp is not as widely used as some other programming languages, it continues to have a dedicated community of developers who value its unique features and capabilities. Nowadays, Lisp has several dialects available, such as Clojure, Scheme, Racket, and Common Lisp. Common Lisp is presently one of the most popular dialects of Lisp, and SBCL (Steel Bank Common Lisp) is the most commonly used implementation of it. SBCL is an open source implementation known for its quick performance and high level of conformity to the Common Lisp standard. To install SBCL on Ubuntu, run the command, '*sudo apt-get install sbcl*' in your terminal. Now, consider the Lisp program '*add.lisp*', shown below, which adds two numbers. Figure 1 shows the execution of this Lisp program. Notice that executing the instruction '*sbcl*' initiates the operation of the SBCL compiler.

```
(format t "Program Loaded")
(defun add (a b)
  (+ a b))
```

Haskell, a true functional programming language, and Scala, which provides many functional programming capabilities, are also frequently utilised for creating AI and